

CH365 Chemical Engineering Thermodynamics

Lesson 17

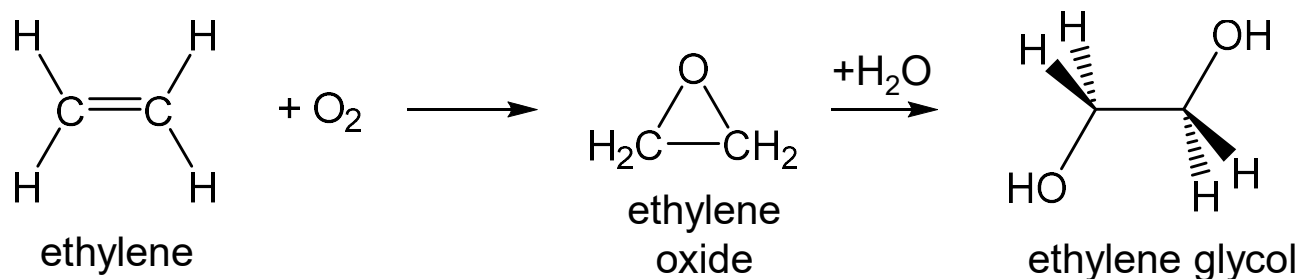
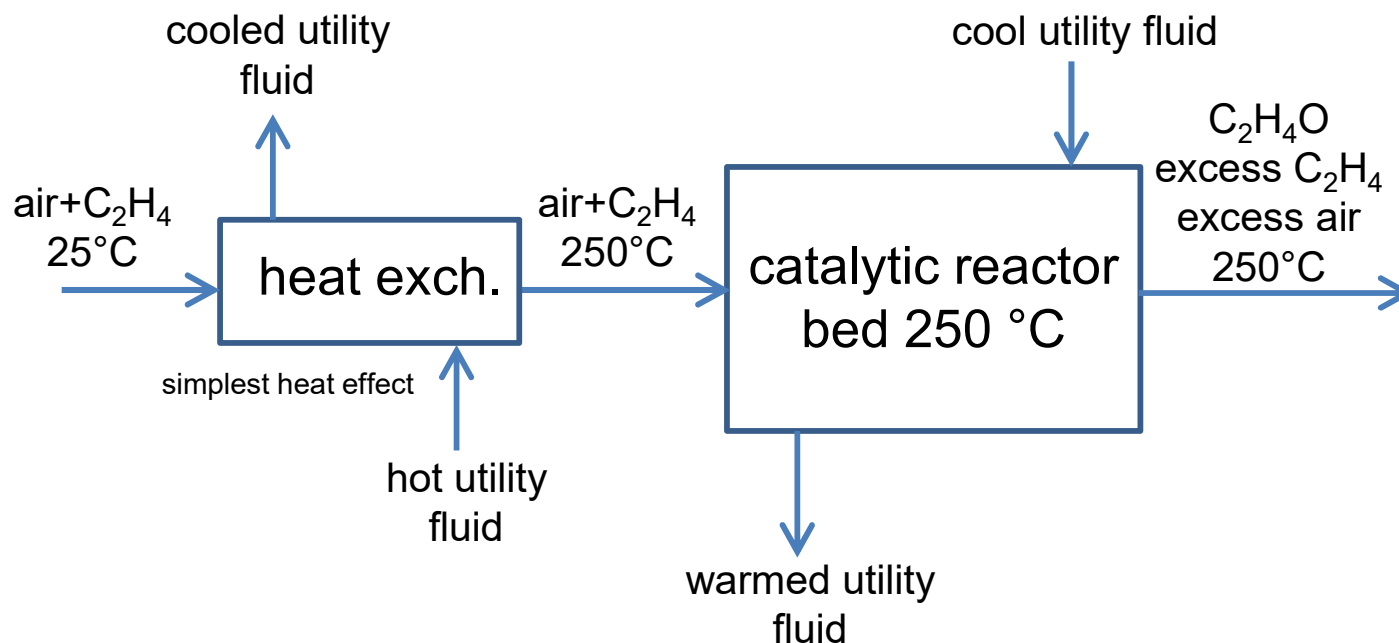
Latent Heat and Heats of Reaction,
Formation, and Combustion

Ethylene Glycol Process

Slide 2

Sensible heat effects – temperature change

➡ Latent heat effects – no temperature change



Latent Heats of Pure Substances

One component evaporating or condensing – No temperature change but heat is transferred. Latent heat of fusion or vaporization.

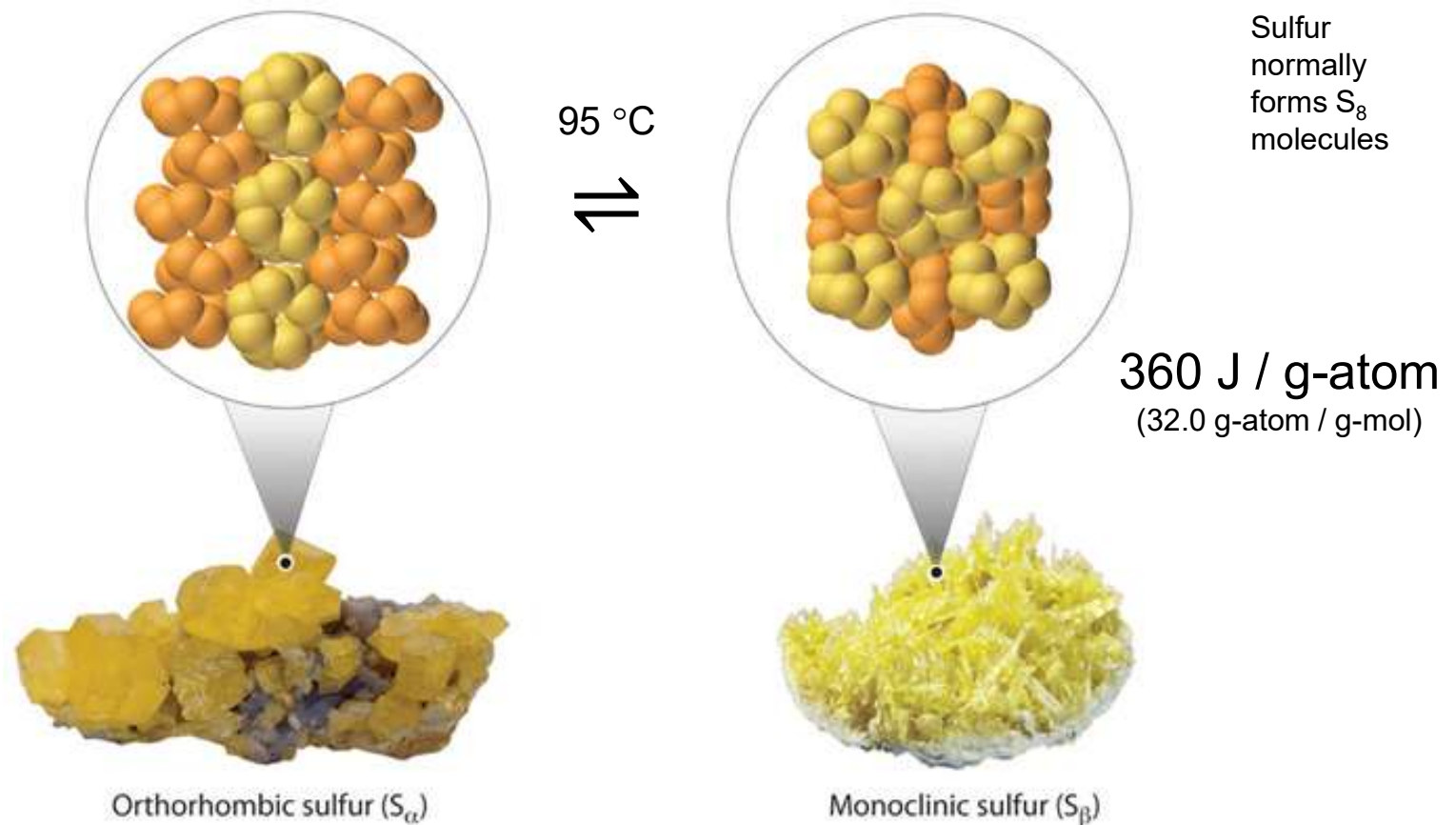


image from <http://2012books.lardbucket.org/books/principles-of-general-chemistry-v1.0>

Latent Heats of Pure Substances

One component evaporating or condensing.

Clapeyron Equation

Calculating ΔH :

$$\Delta H = T \Delta V \frac{dP^{\text{sat}}}{dT}$$

Eq. 4.12, page 146

Molar volume difference
Cubic EOS

P^{sat} given by Antoine Equation

$$P^{\text{sat}} = 10^{A - \frac{B}{T+C}}$$

- exact relationship
- leads to Antoine Eq
- derived in Ch. 6

Rough estimate for
calculating ΔH :

Trouton's Rule:

$$\frac{\Delta H}{RT_n} \sim 10$$

un-numbered equation, p. 142

Rule of Thumb
Errors ~10%

Normal boiling point

Latent Heats of Pure Substances

One component evaporating or condensing.

Reidel's Equation

$$\frac{\Delta H}{RT_n} = \frac{1.092(\ln(P_c) - 1.013)}{0.93 - T_{r_n}}$$

Eq. 4.13, page 147

Empirical Correlation
Errors < 5%

reduced boiling point

Watson's Equation

$$\frac{\Delta H_2}{\Delta H_1} = \left(\frac{1 - T_{r_2}}{1 - T_{r_1}} \right)^{0.38}$$

Eq. 4.14, page 147

Error < 5%

reduced temperatures

Standard Heat of Reaction

Heat effects in chemical reactions are manifestations of the differences in molecular structure.

There are a large number of chemical reactions, each of which can be carried out at an essentially infinite set of process conditions.

Tabulation of all possible heat effects for all possible reactions is impossible.

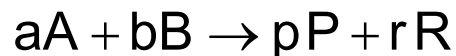
Heats are calculated from data for reactions conducted under standard conditions.

Heat associated with a chemical reaction depends on the temperatures of the reactants and products. A consistent standard treatment is that products and reactants are at the same temperature.

Standard Heat of Reaction

Heats are calculated from data for reactions conducted under standard conditions.

Products and reactants are at the same temperature



Standard Heat of Reaction is the enthalpy change when a moles of A and b moles of B in their standard states at temperature T react to form p moles of P and r moles of R in their standard states at temperature T .

Standard State Definitions:

Gases: ideal gas at 1 bar

Liquids and Solids: real pure liquid or solid at 1 bar

Degree symbol designates standard state

$C_P^\circ = C_P^{\text{ig}}$ for gases

Subscript “f” designates “formation.”

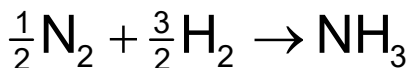
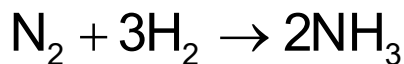


Table C.4, page 672

$$\Delta H_{f,298}^\circ = -46,110 \text{ J}$$



$$\Delta H_{298}^\circ = -92,220 \text{ J}$$

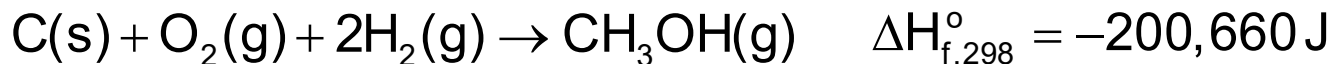
ΔH_{298}°

Standard value at 298.15 K

Standard state properties are functions of T

Standard Heat of Formation

In a *formation reaction*, a compound forms from its constituent elements.

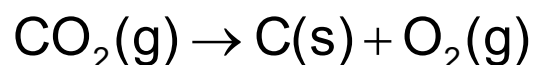


The standard-state heat of formation is based on formation of one mole of a substance from its constituent elements in their standard states and is designated as $\Delta H_{\text{f},298}^{\circ}$.

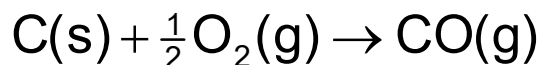
The standard heat of reaction can be calculated if the heats of formation of the reactants and products are known.

Water-gas-shift reaction:

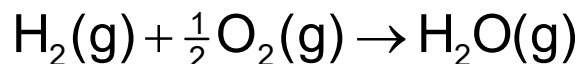
Table C.4, page 672



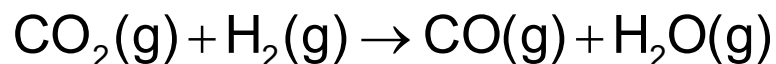
$$\Delta H_{298}^{\circ} = 393,509 \text{ J} \quad \text{negative in Table C.4 (listed as product)}$$



$$\Delta H_{298}^{\circ} = -110,525 \text{ J}$$



$$\Delta H_{298}^{\circ} = -241,818 \text{ J}$$



$$\Delta H_{298}^{\circ} = 41,166 \text{ J}$$

Questions?